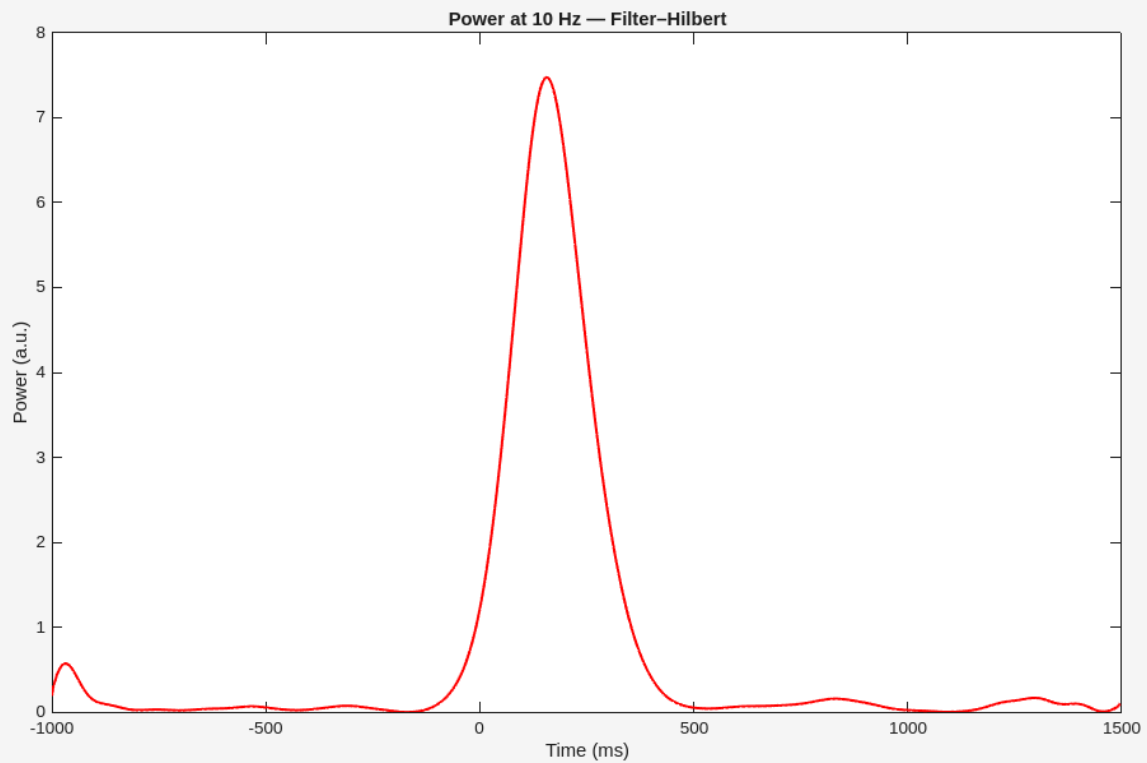
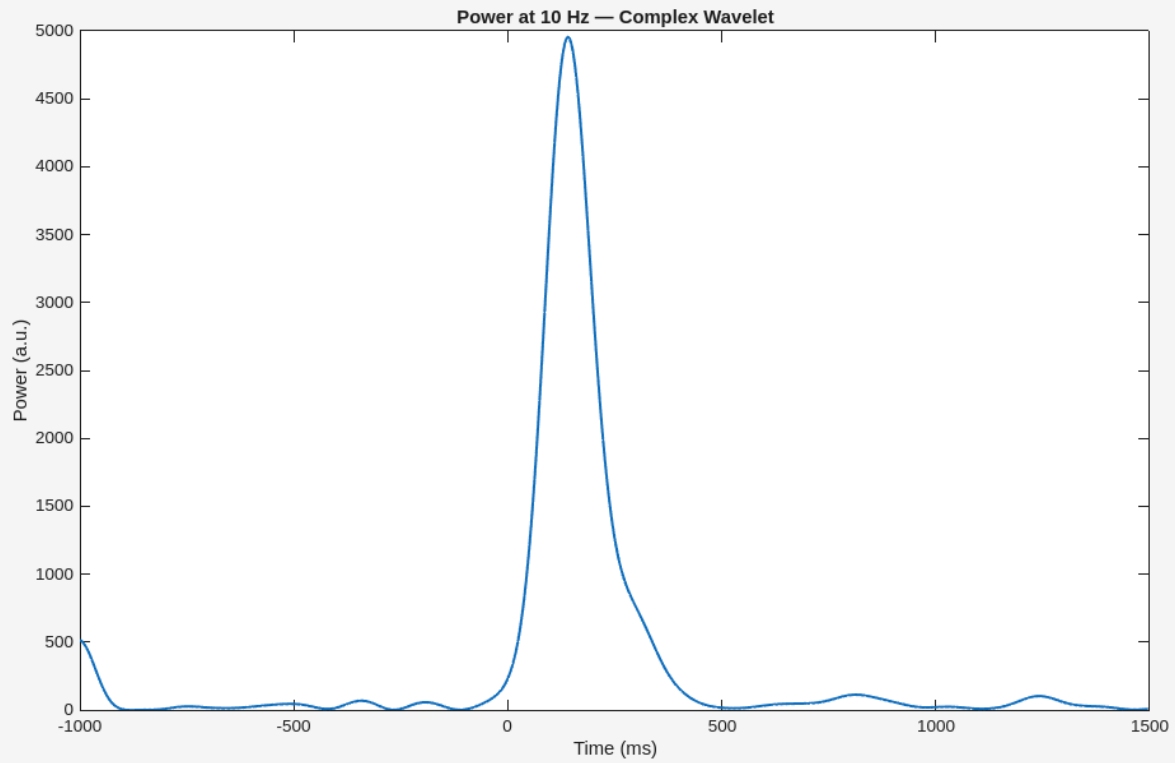
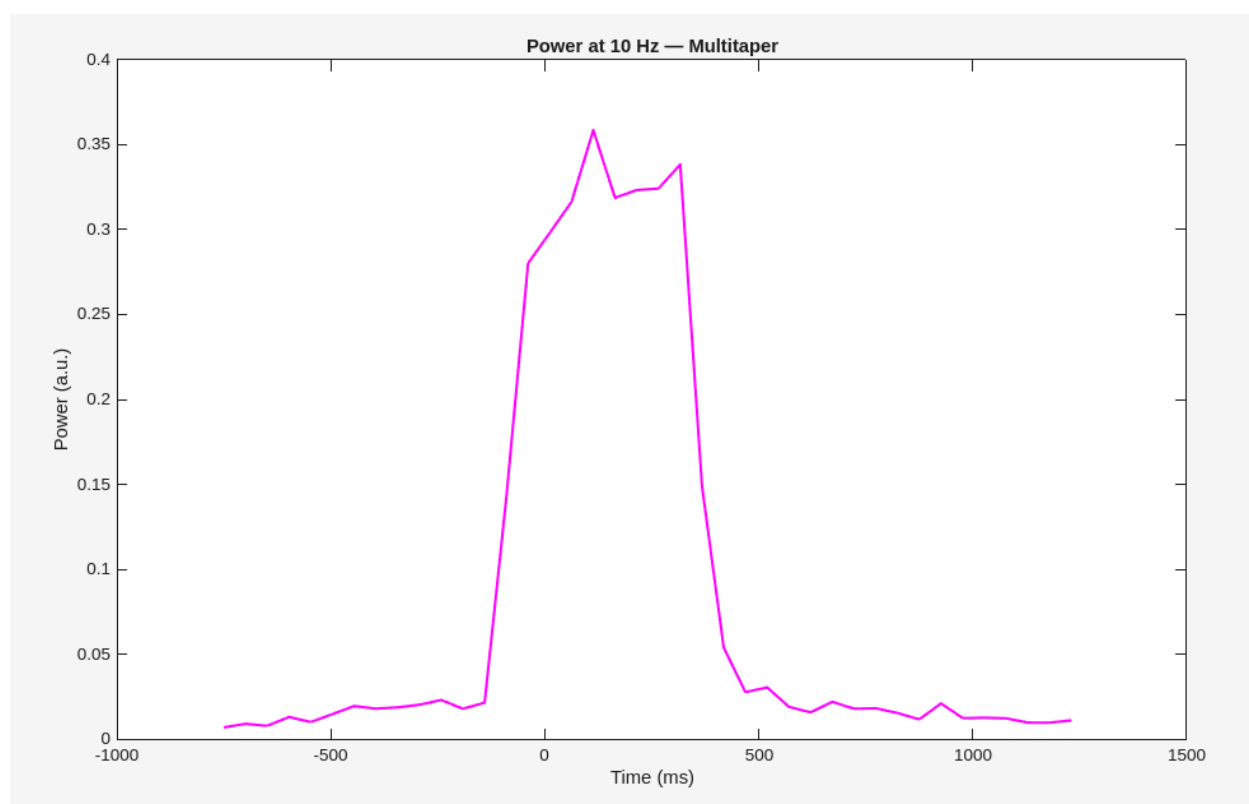
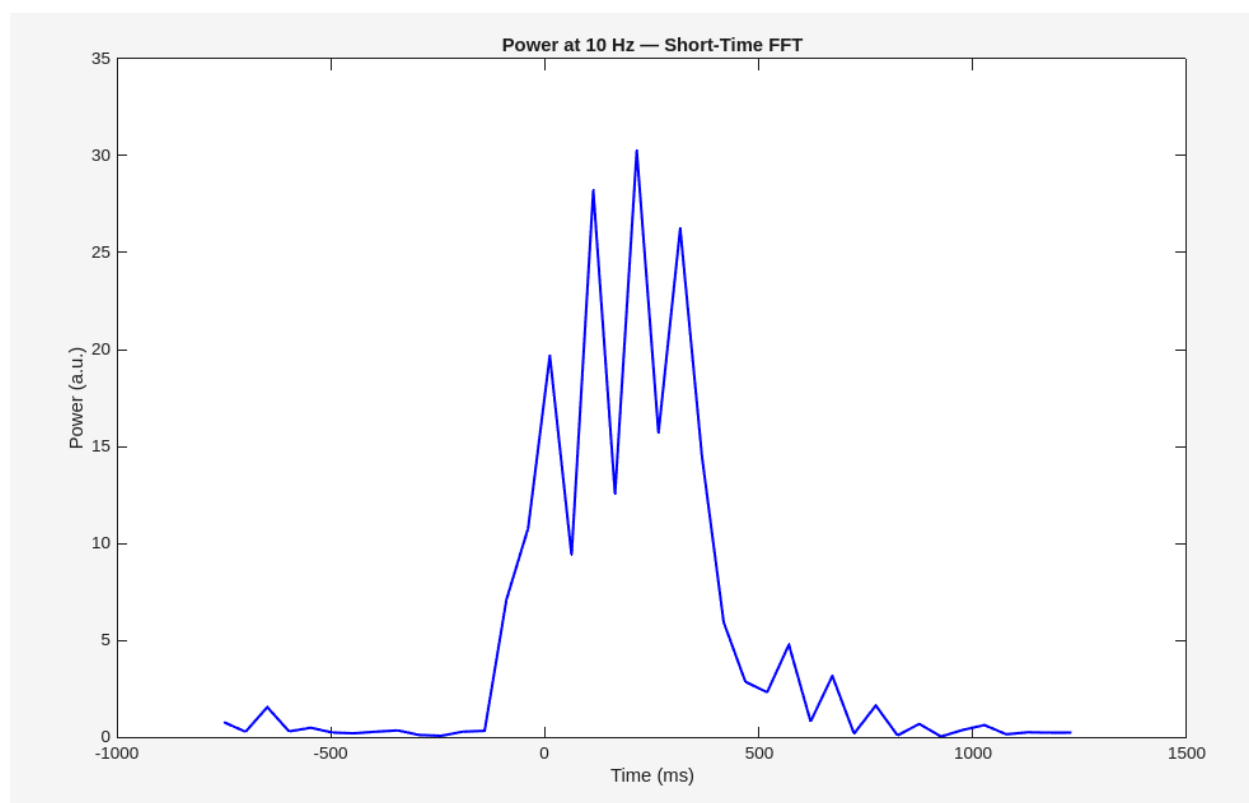
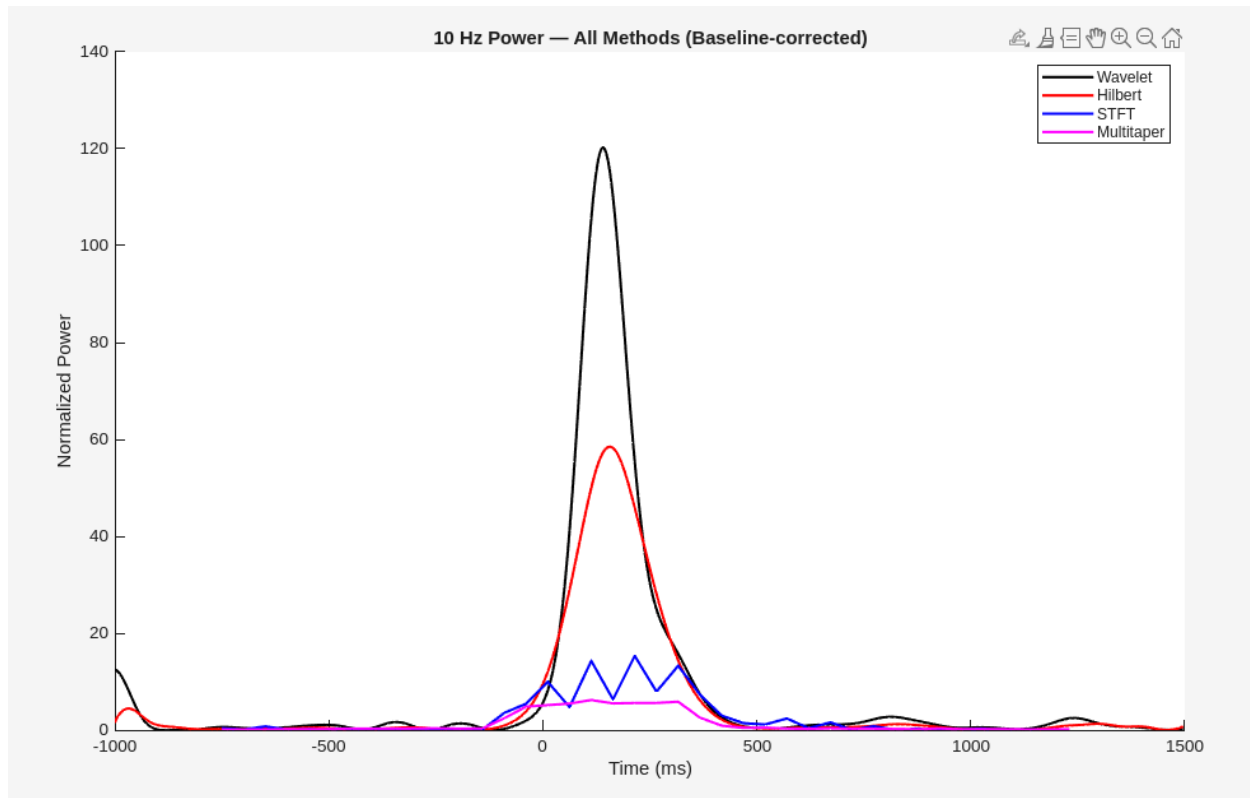


RESULTS AND INTERPRETATION







Interpretation of results:

All four methods show that the power at **10 Hz increases sharply around time 0 ms**. This means that after the event (like a stimulus or signal start), the brain activity around 10 Hz becomes stronger for a short time, and then slowly goes back to normal. So, all methods are detecting the same main effect.

Wavelet Method

- Shows a very strong and sharp power peak right at the event time.
- It reacts quickly, so it shows changes that happen in short time windows very clearly.
- But the power values look very high, because this method's numbers depend on how it scales the signal.
- Best when you want to see fast changes in time.

Filter–Hilbert Method

- Gives a smooth curve with a clear but smaller peak.
- It looks cleaner and less noisy than wavelet.
- This method focuses only on one frequency band (here 10 Hz).

- Best for tracking how power in one band changes over time.

Short-Time FFT (STFT)

- The power looks **rough and step-like**.
- This happens because the FFT splits the signal into time windows, so it can't show very fast changes.
- The peak is still around the same time as other methods, but it looks choppy.
- Works well for **longer, slower changes** in power.'

Multitaper Method

- The curve is smooth and wide, with a smaller peak.
- It gives very stable results and removes random noise well.
- The trade-off is that it shows slower, less sharp changes.
- Good when you want reliable average power instead of fine time detail.

Baseline Correction

The baseline correction makes the plots show how much power changed compared to before the event.

- Wavelet and Hilbert show bigger increases after time 0.
 - STFT and Multitaper show smaller but steadier changes.
- This difference comes from how each method handles time and frequency.

All methods find the same main event — a burst of 10 Hz power after time 0.

But they show it in slightly different ways:

- **Wavelet:** strong and sharp
- **Hilbert:** smooth and clear
- **STFT:** rough and step-like
- **Multitaper:** smooth and steady